

The Motivated Sign in Arabic: Iconicity across Morphology and Grammar

Abstract

This article challenges the Saussurean arbitrariness principle by demonstrating that the Arabic linguistic sign is systematically motivated, and that the relationship between phonological form and meaning in Arabic is non-arbitrary at multiple levels. Drawing on evidence from four interconnected domains, we advance a theoretical framework of phonosemantic iconicity in which articulatory and acoustic properties iconically pre-figure semantic content. Through a qualitative description of morphophonological features, we first demonstrate the diagrammatic iconicity of the Arabic root-and-pattern verb paradigm, wherein the phonological operations of gemination, vowel extension, and affixation enact the semantic operations they encode. Second, we show that the {m-}prefix nominalization system is grounded in the phonetic properties of the bilabial nasal [m], where its cross-linguistic association with embodied presence – evidenced by its near-universal occurrence in ‘mother-words’ – supports its role as a concretizing nominal prefix in Arabic. Third, we examine a set of Arabic grammatical particles, like على > علا ‘on’, that show transparent formal and semantic correspondence with related lexical roots. Finally, we present the analysis of a minimal pair, /q/ ق and /y/ غ, as verb-initial consonants, demonstrating that contrastive phonetic gestures encode contrastive semantic matrices in a structured iconic opposition. Together, these four domains suggest that Arabic constitutes an unusually rich case of systematic iconic motivation in morphology and grammar, posing a significant challenge to strong interpretations of the Saussurean arbitrariness principle.

Keywords: Arabic morphology; Iconicity; Motivated sign; Correspondence; Nominalization.

العلامة المُعلَّلة في اللغة العربية: الأيقونية عبر الصرف والنحو

الملخص:

يقدم هذا البحث تحدياً لمبدأ سوسير "اعتباطية العلامة اللغوية"، وذلك من خلال الإثبات أن الصرف العربي يشكل نظاماً متعدد المستويات من التعليل الأيقوني، تتحدد فيه العلاقة بين الشكل الفونولوجي (الصوتي) والدلالة بنحو نظامي غير اعتباطي. وبالاعتماد على شواهد مستمدة من أربعة محاور مترابطة، نبرهن على أن الأيقونية في اللغة العربية لا تعمل مجرد عمل محصور في مستوى المفردات المعجمية المعزولة، بل تمثل مبدأً تنظيمياً للنظام الصرفي والنحوي برمته. أولاً، نبين أن صيغ الأفعال القائمة على نظام الجذر والوزن في العربية هي صيغ أيقونية تخطيطية؛ حيث تجسد العمليات الفونولوجية – مثل التضعيف، ومد الصوائت، والسوابق، والدواخل – العمليات الدلالية التي تعبر عنها أيقونيا. ثانياً، نبين أن نظام الاشتقاق الاسمي بالسابقة {m} يتأسس على الأيقونية النطقية للصوت الأنفي الشفوي /m/. ثالثاً، نفحص مجموعة من الأدوات والنواسخ النحوية في العربية، والتي تظهر تناظراً شكلياً ودلالياً شفافاً مع الجذور المعجمية المتصلة بها. وأخيراً، نقدم تحليلاً لثنائي أدنى، يتمثل في صوتي القاف [q] والغين [ɣ] بوضعهما في بداية الفعل، مبينين كيف تشفر الحركات النطقية التقابلية مصفوفات دلالية تقابلية ضمن تعارض أيقوني بنيوي. وتشير هذه المحاور الأربعة إلى أن اللغة العربية تشكل حالة نموذجية غنية بشكل استثنائي للتعليل الأيقوني النظامي في الصرف والنحو، مما يضع التفسيرات الراديكالية لمبدأ سوسير حول الاعتباطية أمام تحدٍ جوهري.

الكلمات المفتاحية: الصرف العربي؛ الأيقونية؛ العلامة المُحفَّزة؛ التناظر؛ الاسمية.

1. Introduction

Sound symbolism can be perceived in various instances of words in languages as certain sounds are subconsciously associated with specific objects, actions or concepts. Research shows that even non-sense words like *maluma* and *takete* (Köhler, 1929), are, across cultures, mentally linked to round and spiky shapes, respectively¹. The words ‘spiky’ and ‘round’, and their Arabic equivalents *ṣāʾik* شائك and *mudawwar* مُدَوَّر suggest angular vs. rounded objects, an observation that invites cross-linguistic investigation, even among people unfamiliar with either language. The word for the organ of smell is *nose* in English, *nez* in French and *naríz* in Spanish. The nasal consonant [n] is also present in Arabic *ʾanf* أنف [ʔanf]. Such convergence across these unrelated linguistic traditions (Germanic, Romance, and Semitic) is not coincidental, nor is it the result of borrowing or common descent. It reflects something more fundamental: the nose is the primary organ of nasal resonance, and speakers across unrelated languages have named it using the very sound it produces. The sign, in this case, is not arbitrary; it is motivated.

This observation, which is simple, cross-linguistically verifiable, stands as a direct challenge to one of the most influential claims in the history of modern linguistics: Ferdinand de Saussure's principle of arbitrariness. In his *Cours de Linguistique Générale* (1916), he argued that the relationship between the ‘signifier’, the phonological form, and the ‘signified’, the conceptual content, is fundamentally conventional and unmotivated. No property of the sound sequence [ʃa] *chat* in French determines or reflects the meaning ‘cat’; the association is purely a matter of social convention. This view became axiomatic in 20th century linguistics, shaping structuralism, generative grammar, and much of the theoretical tradition that followed.

Yet, the ‘nose-words’ suggest otherwise. Arabic, the focus of the present study, suggests this facial organ with particular force and systematicity. Consider a preliminary illustration. The Arabic word for *mouth*, *fam* فم [fam], consists of exactly two consonants: [f], a labiodental fricative produced by air escaping through partially open lips, and [m], a bilabial nasal produced by completely sealed lips with continuous nasal resonance. Together, these two consonants enact the mouth's own two fundamental states: open and releasing [f], closed and resonating [m]. The word does not merely refer to the mouth. In its very articulation, it performs it.

1.1 The Problem

The Arabic language presents the linguist with an architecture that is organized at multiple simultaneous levels by iconic motivation, a non-arbitrary relationship between phonological form and meaning grounded in the articulatory and/or acoustic properties of sounds and the embodied experience of producing them. This motivation is not confined to a handful of expressive or onomatopoeic words at the periphery of the lexicon. It penetrates the core of the morphological system – the verb form paradigm, the derivational prefix system, the consonantal root inventory – and extends beyond the lexicon into the grammatical system itself,

¹ The phenomenon was observed earlier by psychologist Uznadze, D. (1924) and later termed as the *Bouba/Kiki* effect by Ramachandran and Hubbard (2001).

shaping the form and function of grammaticalized particles and prepositions. Despite the richness of Arab grammar scholarship (Ibn-Jinnī, d. 1002; Al-Jurjānī, d. 1078; Al-Suyūṭī, 1505) and the research undertaken by contemporary linguists on sound symbolism (Jespersen, 1922; Jakobson, 1960; Bohas, 1997; Magnus, 2001; etc.), the systematic iconic motivation of the Arabic linguistic sign has not, to our knowledge, been comprehensively theorized. The present study attempts to address this gap by proposing a unified framework within which iconic motivation is shown to operate at four hierarchical levels, from the embodied ground of body part vocabulary through the morphological system to the grammatical particle inventory.

1.2 Existing Work

The arbitrariness thesis has never gone entirely unchallenged. Jespersen (1922) documented systematic sound-meaning correspondences in European languages, and Jakobson (1960) argued for iconicity as a fundamental semiotic principle that operates across linguistic levels. Recently, Hinton, Nichols and Ohala (1994) gathered substantial cross-linguistic evidence for *phonosemantic* motivation across unrelated language families, while Magnus (2001) documented systematic consonantal clustering around semantic fields in English. These studies establish that iconic motivation is a genuine, widespread linguistic phenomenon, but none penetrates the internal morphological architecture at the level of detail in the system as Arabic does.

Within Arabic and Semitic linguistics, the question of the relationship between sound and meaning has a long and notable history. The classical Arab grammarians were not oblivious of phonosemantic phenomena; several anticipated aspects of the present framework with remarkable intuitive precision. Ibn Jinnī (ابن جني, d. 1002) in his landmark *Al-Khaṣā'is* (الخصائص) proposed the concept of *al-ishtiqaq al-akbar* (الاشتقاق الأكبر, 'the greater derivation'): roots sharing the same consonants in any permutation share a common semantic field, suggesting that individual consonants carry semantic weight independently of their position². This is a view that encodes the fundamental insight that consonants are not semantically neutral. Al-Jurjānī (الجرجاني, d. 1078) in his *Dalā'il al-I'jāz* (دلائل الإعجاز) developed a sophisticated theory of linguistic motivation in the context of Quranic inimitability, arguing that the relationship between Arabic sounds and meanings reflects a deeper order that cannot be reduced to mere convention. Al-Suyūṭī (السَيُوطِي, d. 1505) similarly documented phonosemantic phenomena in his encyclopedic *Al-Muzhir* (المزهر), noting systematic relationships between articulatory properties and semantic fields.

These classical insights, however intuitively formulated, were never developed into a systematic theoretical framework. The most sustained modern attempt to do so is Bohas's '*Matrix-Etymon theory*' (1997), which proposes that Arabic triconsonantal roots are organized around binary matrices of phonological features carrying stable semantic content. His framework correctly identifies sub-root semantic organization and establishes that individual consonants are semantically non-arbitrary.

² The three 'letters' of a verb-root often share a core meaning when swapped in whatever order (e.g., √k-l-m).

The present study builds on this foundation while departing from it in two important respects. First, the analysis is grounded in *articulatory iconicity* – the physical properties of consonant production – rather than abstract phonological features, making the iconic motivation empirically more transparent and cognitively more associated with embodied experience. Second, and most significantly, the investigation is extended beyond the lexicon into the grammatical particle system, showing that iconic motivation is not confined to the root inventory but shapes the formal and semantic properties of grammatical particles through transparent motivated correspondence with related lexical roots, an extension that has not previously been proposed in the literature, to our knowledge.

Our epistemological stance is explicitly *functionalist-cognitive*, following the tradition of Lakoff and Johnson (1999) in treating linguistic structure as motivated by embodied experience rather than as an autonomous formal system. Within this framework, iconic motivation is not a peripheral phenomenon but a central organizing principle whose scope and systematicity vary across languages and morphological architectures.

1.3 Research Questions and Hypotheses

This study is guided by two research questions:

RQ1: To what extent is the Arabic morphological system iconically motivated (at simultaneous levels, individual consonants, verb form paradigm and grammaticalized particles)?

RQ2: Do Arabic grammatical particles show iconic motivation related to their lexical roots? And if so, does this represent a different pattern from standard grammaticalization, one characterized by transparent motivated correspondence rather than semantic bleaching?

Corresponding to these questions, we advance two hypotheses:

H1: Arabic morphology constitutes a multi-level iconic system in which phonological form and meaning are non-arbitrarily related at the levels of the individual consonant, the triconsonantal root, the verb form paradigm, the derivational prefix system, and the grammatical particle inventory.

H2: Arabic grammatical particles show transparent formal and semantic correspondence with related lexical roots – a correspondence grounded in articulatory iconicity that persists synchronically without semantic bleaching. This constitutes a pattern distinct from standard grammaticalization, suggesting that the Saussurean arbitrariness thesis cannot be maintained as a universal principle of human language.

1.4 Methodology

The methodology in this study is qualitative and analytical, appropriate to its theoretical aims. It draws on four specific analytical procedures:

Phonetic gesture analysis: For each consonant examined, a precise articulatory description is provided – place of articulation, manner of articulation, voicing, and airstream properties – which serves as the empirical basis for proposing the consonant's semantic matrix.

Semantic field mapping: For each proposed consonantal matrix, a systematic survey of Arabic roots sharing that initial consonant is conducted, identifying semantic clustering, testing the matrix's coverage, and honestly accounting for apparent outliers.

Minimal pair analysis: Consonants sharing place of articulation but differing in manner – specifically /q/ ق and /ɣ/ غ – are compared systematically to demonstrate that contrastive phonetic gestures encode contrastive semantic matrices.

Motivated correspondence analysis: For each case study, the formal and semantic relationship between the grammatical particle and its related lexical root is examined — identifying the shared consonantal matrix, the semantic coherence between lexical and grammatical meanings, and the synchronically transparent nature of the correspondence.

The data consists of classical Arabic lexical material drawn from Ibn Manẓūr's *Lisān al-ʿArab* (لسان العرب) and *Al-Muʿjam al-Wasīṭ* (المعجم الوسيط), grammatical data from classical Arabic grammars and Quranic Arabic, and some dialectal data from Maghrebi Arabic.

1.5 Article Structure

The article proceeds as follows. Section 2 presents the theoretical framework, the iconic cascade, within which all subsequent analyses are situated. Section 3 demonstrates the diagrammatic iconicity of the Arabic verb form paradigm and the {m-} prefix nominalization system. Section 4 examines three cases of motivated correspondence between Arabic grammatical particles and related lexical roots, showing that articulatory iconic motivation extends from the lexicon into the grammatical system. Section 5 presents a minimal pair analysis of the consonants /q/ and /ɣ/, extended through the cosmological opposition of East/West, غرب / شرق. Section 6 draws theoretical implications and concludes.

2. Theoretical Framework: The Iconic Cascade

2.1 The Gradient Model of Motivation

Rather than treating arbitrariness and iconicity as a binary opposition, this study adopts a *gradient model of motivation* in which linguistic signs range along a continuum from maximally arbitrary to maximally iconic. Most signs occupy intermediate positions, partially motivated, partially conventional. The critical claim is not that all Arabic signs are maximally iconic, but that the root-and-pattern morphological architecture of Arabic systematically positions its signs toward the iconic end of this continuum, more consistently and more deeply than the concatenative morphologies of languages such as English, French, or Spanish.

The non-concatenative structure of Arabic is crucial here: consonantal roots remain visible and analytically separable across derivation, preserving iconic motivation in ways that concatenative morphology tends to obscure through phonological fusion and erosion. A French speaker cannot isolate a root consonant carrying a semantic matrix from a derived form; the iconic motivation, if present, is buried under layers of affixation and phonological change. An Arabic speaker, by contrast, can always identify the triconsonantal skeleton and recover its iconic content. The architecture *protects its own iconicity*.

2.2 The Iconic Cascade — Four Levels

We propose that iconic motivation in Arabic operates as a *cascade*, a vertically organized system in which motivation flows from the most embodied and concrete level to the most abstract and grammatical. Four levels are distinguished:

Level 0 — The Body

The foundational level, pre-linguistic and universal. The body is the original site of meaning: before any convention, before any social agreement, the body's experience of itself generates the most fundamental conceptual schemas. Body part nouns exemplify this level most directly: أنف [ʔanf] (nose) and فم [fam] (mouth) name organs with the sounds those organs produce or the movements they make. This level grounds the framework in embodied cognition theory (Lakoff & Johnson 1999) and provides the physiological foundation for the iconic motivation operating at higher levels.

Level I — The Phonetic Gesture

A phonetic gesture is the specific articulatory act – place of articulation, manner of articulation, voicing, airstream direction – that produces a given consonant. The claim at this level is that phonetic gestures are not semantically neutral: their physical properties iconically pre-figure semantic content before any lexical convention is established. The abrupt uvular closure and release of [q] physically enacts severance and boundary-creation; the continuous voiced uvular friction of [ɣ] physically enacts depth and engulfment; the smooth lateral flow of [l] physically enacts unobstructed directionality. This iconic relationship is physiological before it is cognitive — grounded in the body's experience of producing and perceiving these sounds.

Level II — The Semantic Matrix

A semantic matrix is the abstract conceptual schema that a phonetic gesture encodes — abstract enough to generate a field of related meanings across multiple domains, yet specific enough to make testable predictions about the distribution of meanings across the root lexicon. The matrix of [q], for instance, is not merely "cutting" — it is the broader schema of **decisive boundary-creation**, which generates meanings as varied as physical severance (قطع), mathematical division (قسم), epistemic decision (قَرَر), and divine decree (قَدَر). The semantic matrix is not a gloss but a **conceptual primitive** — an abstract structure instantiated in physical, social, temporal, and epistemic domains.

Level III — The Lexical and Morphological Field

At this level the semantic matrix is instantiated across the triconsonantal root lexicon and the morphological derivational system. In many instances, each root represents a specific instantiation of its initial consonant's matrix, modulated by the second and third consonants. The verb form paradigm operates at this level as a second, orthogonal system of iconic organization – the phonological operations of gemination, vowel extension, prefixation, and infixation iconically encoding relational semantic operations of intensification, reciprocity, reflexivity, and causation respectively.

Level IV — Grammatical Particles

The fourth level concerns Arabic grammatical particles — prepositions, aspectual markers, and clitics — whose formal and semantic properties show transparent correspondence with related lexical roots. This level differs from the preceding three in that it operates at the boundary between the lexicon and grammar. The critical claim at this level is not that lexical items historically shed their referential content and became grammatical operators, a diachronic claim that the Arabic textual record does not always support, but rather that certain grammatical particles share consonantal and semantic properties with related lexical roots in ways that are synchronically transparent and iconically motivated. This claim is developed in Section 2.3.

2.3 Motivated Correspondence Between Lexical and Grammatical Forms

The central theoretical contribution of this study lies in the following principle:

- *Certain Arabic grammatical particles show transparent formal and semantic correspondence with related lexical roots: the consonant carrying the most iconically motivated matrix is shared between lexical and grammatical forms, and the grammatical function is semantically coherent with that matrix.*

This principle has two components:

Formal correspondence: The phonological substance shared between a lexical root and a related grammatical particle is not arbitrary — it is the most iconically motivated element of the root. The particle *على* *ʕalā* shares the verb root *علا* *ʕalā* 's spatial elevation schema; *لِ* and *إلى* *li/ʔilā* share the {l} matrix of smooth unobstructed directionality with *وَلِيّ* *waliya*; *قَدْ* *qad* shares the {q} matrix of decisive boundary-creation with *قَدَر* *qadara*.

Semantic coherence: The grammatical function of each particle is semantically coherent with the matrix of the shared consonant. *على* encodes superposition — being above — coherent with *علا* *ʕalā* 's meaning of rising / being high. *لِ* / *إلى* *li/ʔilā*, encode directionality toward, coherent with *وَلِيّ* *waliya* 's meaning of smooth proximity/following. *قَدْ* *qad* places events at the boundary between possible and real, coherent with *قَدَر* *qadara* 's meaning of measuring and decreeing.

This differs from the standard grammaticalization account in one important respect: we do not claim that the lexical items historically developed into grammatical particles through semantic bleaching. The Arabic textual record, including the Quran, the earliest attested Arabic text, shows these particles already functioning grammatically from the outset, coexisting transparently with their related lexical roots. What we observe is therefore not a historical development but a **synchronically stable iconic correspondence**, grounded in the same articulatory motivation that organizes the consonantal matrix system at Levels I through III.

This also means that the relationship between lexical root and grammatical particle in Arabic is not, as standard accounts would have it, historically recoverable but synchronically opaque. It is both historically stable and synchronically transparent. The motivated correspondence between form and function is not a residue of a forgotten etymology but an active, visible property of the language's grammatical system.

2.4 The Two Axes of Iconic Organization

The iconic cascade operates along two orthogonal axes that together constitute the article's empirical scope:

- **Axis 1: The Paradigmatic Axis (Consonantal matrices)**

This axis concerns the identity of the consonants chosen for a root – their place and manner of articulation and the semantic matrices they carry. It answers the question: “What kind of event or state is being encoded?” The axis is qualitative, specifying the nature of the action through the iconic properties of the initial consonant.

- **Axis 2: The Syntagmatic Axis (Verb form system)**

This axis concerns the phonological operations performed on the root – gemination, vowel lengthening, prefixation, infixation. It answers the question: “How is that event structured in relation to participants?” Here, the axis is relational; it specifies the force, directionality, and reflexivity of the action through diagrammatic iconicity.

These two axes interact productively: a learner who has internalized k consonantal matrices and n verb forms has not acquired $(k + n)$ meanings but $(k \times n)$ meanings – the iconic systems multiply rather than add, making the system *generatively iconic* rather than merely locally motivated. In other words, a learner who has internalized a set of consonantal matrices and a set of verb forms has not acquired the sum of their meanings but their product – the iconic systems multiply rather than add. A single concrete example illustrates this: the consonantal matrix of ق [q] (boundary-creation) combined with Form VII (انفعل *ʔinfaʕala* – spontaneous intransitive) produces انْقَطَعَ *ʔinqaṭaʕa* (to be severed spontaneously) – a meaning that is fully derivable from the intersection of the two iconic systems without independent memorization.

3. Root-and-Pattern Morphology as a Diagrammatic Iconic System

3.1 The Architecture of Arabic Morphology

Arabic morphology is primarily non-concatenative system based on a ‘root-and-pattern’ arrangement: grammatical / derivational information is not encoded through the sequential addition of prefixes and suffixes to an invariant stem, as in English or French, but through the interleaving of a consonantal root with a vocalic pattern (McCarthy 1981). The root $\sqrt{k-t-b}$, for instance, carries the abstract semantic content of ‘writing’, while patterns such as *CaCaC*, *CiCāC*, and *maCCaCa* encode grammatical and derivational meanings, producing كَتَبَ *kataba* (he wrote), كِتَاب *kitāb* (book), and مَكْتَبَة *maktaba* (library). This architecture has two consequences directly relevant to the present study: the consonantal root remains evident and analytically separable, and the vocalic pattern carries grammatical content constant across roots, creating the conditions for a *two-dimensional iconic system* in which root and pattern encode complementary dimensions of meaning.

The present section argues that the Arabic verb form paradigm is *diagrammatically iconic*: the phonological operations applied to the root (gemination, vowel extension, prefixation, and infixation) enact the very semantic operations they encode. The form does not merely signal the meaning; it performs it.

3.2 Verb Forms and Iconic Organization

The common verb forms of Arabic (*ʔawzān* = ‘measures’) the unmarked **Form I** *faʕala*, reveal a subtle analogy of relations and internal logic. The augmented patterns, based mostly on tri-consonantal roots and grounded on the fundamental principle of *an increase in structure indicates an increase in meaning*³, operate along three phonological dimensions – *quantity*, *position*, and *directionality* –, each carrying an iconic motivation.

- The most transparent case is **Form II** *faʕʕala* (فَعَّلَ), where gemination of the middle consonant iconically encodes augmented semantic force and directly enacts intensification, as in *kassara* كَسَّرَ > ‘to smash repeatedly’, or causation, as in *ʕallama*, عَلَّمَ > to teach/cause to know. A learner who grasps this relationship derives both meanings from a single iconic principle rather than memorizing them separately. The phonological act is precise and the iconic mapping is direct: **Gemination in phonological substance => Augmentation of semantic force.**
- **Form III** *fāʕala* (فَاعَلَ) lengthens V1 between C1 and C2 — stretching the root open at its centre. The iconic mapping is precise: **Extension of vowel ==> extension of action.** The long vowel literally occupies the acoustic space between two participants; e.g., *kātaba* كَاتَبَ (Form I كَتَبَ *kataba* ‘to write’) means ‘to correspond with a partner’.

³ A rule that traditional Arab grammarians agreed upon, including Ibn-Jinni (4th H) زيادة المبني تدل على زيادة المعنى

The long open vowel is the *phonological enactment of relational extension*. The sound reaches outward from the subject toward another person, iconically representing the reciprocal action.

- **Form IV** *ʔafʕala* - أَفْعَلَ - adds a glottal stop, the ʔ-prefix before the root, an element inserted from outside the consonantal skeleton, which iconically encodes an external initiation of the verbal process, representing an **external boundary being crossed**; while خَرَجَ *ḫaradʒa* is ‘to go out’, أَخْرَجَ *ʔaḫradʒa* means ‘to cause to exit’.
- **Forms V and VI** *tafaʕʕala* and *tafāʕala*, add a {t} prefix to Forms II and III respectively turning their meanings reflexively back upon the subject, as in تَعَلَّمَ *taʕallama* ‘to teach oneself/to learn’ and تَقَاتَلَ *taqātala* ‘to fight each other’.
- **Form VII** *infaʕala* (انْفَعَلَ) adds a nasal prefix whose interior resonance iconically encodes the inward turning of the action: *ʔinkasara* اِنْكَسَرَ (to be broken spontaneously) — the action received inwardly by the subject rather than imposed from outside.

The other verb forms, too, comprise affixes that carry some iconic motivation encoding a connection to the meaning: the position of the added material – outside, before, or inside the root – maps onto the position of the action relative to the subject: external, passive-internal, or reflexive-internal. The overall architecture of the verb forms illustrates the broad principle:

- ***phonological operations in the Arabic verb paradigm are not arbitrary conventions but iconic enactments of the semantic operations they encode.***

3.3 The {m-} Prefix Nominalization System

The iconic motivation of the Arabic morphological system extends beyond the verb form paradigm into the derivational system of nominal formation. The most productive and most iconically transparent derivational prefix in Arabic is {m-}, the bilabial nasal that appears in five distinct nominal categories, with the triple-vowel modulation: {ma-}- {mu-}-{mi-}, each specifying a given noun category.

Table 3.1 {m-} prefix in nominal categories

Category	Pattern	Example	Transcription	Meaning
اسم المكان	مَفْعَل	مَكْتَب	<i>maktab</i>	place of writing — office
اسم الآلة	مِفْتَاح مَفْعَل/مِفْعَال	مِفْتَاح	<i>miftāḥ</i>	instrument of opening — key
اسم المفعول	مَفْعُول	مَكْتُوب	<i>maktu:b</i>	that which has been written — letter
اسم الفاعل	مُفْعِل	مُعَلِّم	<i>muʕallim</i>	one who teaches — teacher
مَفْعَل المصدر الميمي		مَذْهَب	<i>maʔhab</i>	the act of going — doctrine

What is immediately striking is that a single consonant, [m], unifies five distinct grammatical categories. This is not arbitrary: [m] *carries an iconic motivation powerful enough to serve as the organizing principle of the entire nominal derivation system*. The phonetic profile of the bilabial nasal [m] involves two simultaneous articulatory gestures: *complete oral closure* (both lips sealed) and *continuous nasal resonance* (air flowing through the nasal passage). This dual gesture enacts a fundamental semantic operation:

- ***Complete containment with continuous resonance ==> the bringing of a dynamic process into stable, bounded, concrete existence ==> reification***

The verbal process - dynamic and event-denoting - is given a *body*: a place, *maktab* (مكتب), an instrument, *miftāḥ* (مفتاح), a result, *maktu:b* (مكتوب), an agent, *muṣallim* (معلم), or an abstract noun, *maʿlūm* (مذهب). In every case, {m-} performs the same fundamental operation: *it nominalizes by concretizing*, giving the verbal process a stable, thing-like existence in one of several possible domains. The cross-linguistic evidence is compelling: [m] appears in words for ‘mother’ across unrelated languages — *mama*, *mère*, *madre*, *ʔumm*, a convergence Jakobson (1960) attributes to the nursing infant's bilabial closure while breast-fed. The Arabic {m-} nominalization system is grounded in this universal phonosemantic association. This cross-linguistic universality is significant for the article's argument: it reveals the iconic motivation of the {m-} nominalization system, not specifically in Arabic, but also as grounded in a *universal phonosemantic association* between bilabial nasality and embodied concretization, one of the strongest known cases of phonetic iconicity in language.

The verb form architecture and the {m-} nominalization system together constitute a two-dimensional iconic space:

- Axis 1 — consonantal matrices — answers the question: *What kind of event is encoded?*
- Axis 2 — morphological patterns — answers: *How is that event structured and what nominal entity does it produce?*

The combination is **generative**: A speaker who has internalized the consonantal matrices and the pattern meanings can derive the approximate meaning of an unfamiliar form without memorization. This generativity is the most practical consequence of the iconic architecture, and its most powerful argument against arbitrariness.

4. Grammaticalization or Motivated Correspondence?

4.1 The Principle and Its Scope

The Arabic grammatical particles examined in this section have sometimes been discussed within the framework of *grammaticalization* – the process by which lexical items progressively shed their referential content and acquire grammatical function (Hopper & Traugott 2003; Heine, Claudi & Hünnemeyer 1991). However, the Arabic data considered here does not comfortably fit this account. Standard grammaticalization theory predicts semantic

bleaching, the progressive loss of lexical meaning to the point where the connection between lexical source and grammatical output becomes synchronically opaque. This is not what we observe in Arabic. Rather, the particles behave more like lexical derivations, particularly from verb-root material: they coexist transparently with their related roots, retain clear semantic connection, and show no evidence of bleaching.

The preposition *ʕalā* على 'on', for instance, coexists in the language with the verb *ʕalā/jaʕlū* علا/يعلو, meaning 'to rise, to be high'. The phonological and semantic relationship between the verb علا 'to rise' and the preposition على 'on' is too obvious to be unnoticed: a speaker of Arabic is (sub)consciously aware of the connection, particularly as the pronunciation of the two items is identical. Other interesting examples of co-existence are shown below:

Table 4.1 Particle / Lexical Item relation

Particle/preposition	Lexical item	Meaning
إلى + clitic ل <i>ʕilā/li</i> 'to/towards'	وَلِيّ <i>waliya</i>	to be adjacent/to follow
قَدْ / <i>qad</i> 'two functions'	قَدَرَ / <i>qadara</i>	to measure/to have power
فَوْقَ / <i>fawqa</i> 'on/above/over'	فَاقَ / <i>fāqa</i>	to surpass/to be above
فِي / <i>fī</i> 'in / inside; containment'	فَمَ / <i>fam</i> / فَوْفِي / <i>fū/fī</i>	mouth / opening

Particularly interesting is the *structured opposition* between قَبْلَ and بَعْدَ, 'before and after', formally and semantically corresponding, respectively, with the verbs قَبِلَ *qabila*, 'to receive/to accept' and بَعُدَ *baʕuda*, 'to be distant'. These temporal prepositions appear to encode opposite spatial/temporal schemas inspired, as it were, by the corresponding verbs.

The particle *qad* قَدْ, is an aspectual/modal particle phonologically reduced from the verb قَدَرَ *qadara* (to measure). Just like the verb it is related to, قَدْ belongs to the well-established {q-} matrix of *decisive boundary-creation*. With its two well-documented functions as a verbal construct in Arabic, it indicates a) certainty or completion when preceding a perfective verb (الماضي), as in قَدْ جاء الطبيبُ 'The doctor has come', and b) possibility or probability before an imperfective verb (المضارع), as in قَدْ يَأْتِي الْيَوْمَ 'He may come today'. *The particle قَدْ then places the event at the boundary between the possible and the real.* This is motivational persistence in its clearest form: the referential content of the verb قَدَرَ – measuring, having power, decreeing – persists in the particle *qad* قَدْ with the {q-}-matrix and its iconic motivation. The unifying schema is the {q-} matrix: *decisive boundary-creation* – the imposition of a limit or threshold on what is, what can be, or what will be. This is *formal persistence*: the phonological reduction in *qad* is not random; it preferentially retains the most iconically motivated element of the source form, the [q] consonant, and operates at a more abstract level than the full lexical قَدَرَ.

In every case both items are alive in the language simultaneously, semantically related and formally transparent. What the Arabic data suggests, therefore, is not grammaticalization in the standard sense but rather a pattern of motivated lexical correspondence – grammatical particles that are transparently related to lexical roots through shared consonantal matrices and

coherent semantic fields. This pattern is grounded in the same articulatory iconicity that organizes the root lexicon, and constitutes further evidence for the motivated sign in Arabic.

Therefore, rather than tracing diachronic development pathways, which the Arabic textual record does not always allow⁴, the analysis focusses on the synchronically observable connection between the particle and the root, a relationship that is psycholinguistically real, actively alive in the speaker's mind. The few cases examined illustrate similar types of correspondence between lexical items and grammatical forms, sharing the consonant(s) carrying the most iconically motivated matrix with the related lexical roots, and the grammatical function is semantically coherent with that matrix.

As for the complex وَلِيّ [waliya] → إِلَى [ʔilā] → لِ [li], the unifying schema is the **{l-} matrix** of smooth, unobstructed continuity, the lateral approximant [l] whose articulatory profile can be characterized as: **smooth, unobstructed flow, encoding proximity, and directionality**

- √w-l-j is a weak root whose semi-vowels و (wāw) and final ي (jāʔ) make the root particularly susceptible to reduction, and what survives across all three stages is ل, the only fully stable, fully consonantal element of the root, carrying the {l-} matrix intact.

The semantic pathway from the verb وَلِيّ to the prepositions إِلَى / لِ follows a universal process: **spatial proximity** → **directionality** → **purpose** → **possession**, documented cross-linguistically by Heine *et al.* (1991). What is distinctive about the Arabic case is that this pathway is *iconically grounded*: the {l-} matrix of smooth unobstructed flow motivates both the source (smooth adjacency) and the output (smooth directionality).

4.2 Motivational Persistence — A Synthesis

The cases examined in this section demonstrate motivated formal and semantic correspondence between Arabic grammatical particles and related lexical roots. Several observations emerge:

- First, in each case the particle shares the consonant carrying the most iconically motivated matrix with its related root — [q] in *qad*, [l] in *ʔilā/li*, [f] in *fī*...
- Second, the grammatical function of each particle is semantically coherent with the matrix of the shared consonant, not merely historically connected to it.
- Third, the correspondence is transparent and synchronically observable; it does not depend on reconstructed historical development but on the iconic relationship between articulatory properties and semantic content.
- Together these observations constitute evidence for motivated correspondence between lexical and grammatical form, grounded in the same articulatory iconicity that organizes the root lexicon.

⁴ Unlike Romance languages, *Al-Lughā / Fuṣṣḥā*, referred to as Classical Arabic, was codified and frozen by early Arab grammarians (7th / 8th C.) to preserve the language of the *Qurʾān*; the standard masks its evolution.

5. Consonantal Matrices — Analysis of a Minimal Pair

5.1 Sub-Root Iconicity — Individual Consonants as Semantic Determinants

The analyses presented in Sections 3 and 4 have operated primarily at the level of morphological patterns and iconicity pathways. The present section descends to a deeper level of the iconic cascade — Level I and Level II — where individual consonants, through their articulatory properties, encode stable semantic matrices that organize the root lexicon into coherent semantic fields.

The claim that individual consonants carry semantic weight is not new. Ibn Jinnī's concept of *al-ishtiqaq al-akbar* (الاشتقاق الأكبر), the greater derivation, 10th C.) proposed that roots sharing the same consonants in any permutation may share a common semantic field, implying that consonants are semantically non-arbitrary independently of their position. Bohas' Matrix-Etymon theory (1997) developed this insight formally, proposing binary feature matrices as the organizing principle of root semantics. The present study departs from both predecessors in grounding the analysis in *articulatory iconicity*, the physical properties of consonant production, rather than in permutational relationships or abstract phonological features.

The methodological approach at this level involves three steps: a precise articulatory description is provided for each consonant; a semantic matrix is proposed; the matrix is tested against the root lexicon. The primary evidence presented in this section takes the form of a *minimal pair analysis*, the most rigorous possible test of the consonantal matrix hypothesis. Two consonants, [q] and [ɣ], sharing the same place of articulation but differing in manner are compared: if contrastive phonetic gestures encode contrastive semantic matrices, then this constitutes strong evidence that the iconic motivation is systematic rather than coincidental.

5.2 The ق [q] Matrix — Decisive Boundary-Creation

5.2.1 Articulatory Profile and Semantic Matrix

The consonant ق [q] is a voiceless uvular plosive whose articulatory gesture is physically decisive and abrupt: complete occlusion followed by sudden, forceful release. Nothing in the vocal tract produces a more complete interruption of the airstream, and nothing produces a more abrupt termination of that interruption. The phonetic gesture physically enacts *complete severance followed by decisive release*, a boundary created and crossed in a single articulatory act. From this articulatory profile we derive the following semantic matrix:

- ***Decisive boundary-creation - the abrupt, forceful demarcation of one state, entity, or proposition from another***

This matrix is abstract enough to generate meanings across multiple domains — physical, social, epistemic, and theological — while specific enough to make testable predictions about the distribution of meanings across [q]-initial roots.

5.2.2 Lexical Evidence of the [q]-matrix

The following roots illustrate the [q]-matrix across diverse semantic domains:

Table 5.1 [q]-matrix Physical domain

Root	Meaning	Matrix instantiation
قطع	to cut, sever	paradigm case — physical boundary between two pieces
قسم	to divide, partition	boundary between portions
قصّ	to cut, trim	precise boundary-creation
قضم	to break, shatter	forceful boundary through matter
قلع	to uproot, extract	decisive rupture between entity and ground
قدّ	to split lengthwise	boundary along an axis
قطّ	to cut crosswise	boundary across an axis

Table 5.2 [q]-matrix Social/epistemic domain

Root	Meaning	Matrix instantiation
قرّر	to decide	boundary between deliberation and commitment
قبل	to accept	boundary crossed from refusal to consent
قهر	to overpower	boundary of domination imposed on another
قضى	to judge, decree	boundary between right and wrong, guilty and innocent

Table 5.3 [q]-matrix Theological / Ontological domain

Root	Meaning	Matrix instantiation
قدر	to measure, to decree	boundary imposed on what is and what can be
القدر	fate, divine decree	the ultimate boundary on events
قام	to rise, to undertake	transition across a positional boundary

The semantic clustering is robust across domains all instantiating the same abstract schema:

- *the creation of a threshold that separates one state from another.*

5.2.3 Assessment of Outliers

Not all [q]-initial roots fall straightforwardly within the proposed matrix. Roots such as قال *qāla* (to say), قام *qāma* (to stand/rise), and قلب *qalaba* (to turn/flip) require a more abstract reading. However, as argued in Section 2.2, the semantic matrix must be understood at a level of abstraction sufficient to generate diverse instantiations:

- قال — speech as a decisive act of demarcation — cutting through silence with a bounded, delimited utterance
- قام — rising as a transition across a positional boundary — from lying to standing, from inaction to action
- قلب — flipping as a categorical reversal — crossing entirely from one state to its opposite

These readings are not forced: they are consistent with the matrix and with the cognitive linguistic principle that abstract schemas are instantiated across multiple concrete domains (Lakoff & Johnson 1999). Nevertheless, we acknowledge that the coverage of the matrix is not exhaustive; some [q]-initial roots resist straightforward iconic motivation, and this honest acknowledgment is itself part of the framework's falsifiability.

5.3 The غ [ɣ] Matrix — Engulfment into Depth

5.3.1 Articulatory Profile

The consonant غ [ɣ] is a *voiced uvular fricative*, whose articulatory and acoustic properties distinguish it sharply from ق [q] despite their shared place of articulation:

- **Place:** Uvular — identical to ق, produced at the rearmost point of the oral cavity
- **Manner:** Fricative, the back of the tongue approximates but does not fully contact the uvula, producing continuous turbulent airflow rather than complete closure and release
- **Voicing:** Voiced — the vocal cords vibrate continuously throughout, adding resonant depth to the turbulence
- **Acoustic character:** Unlike the sharp, discrete burst of [q], [ɣ] produces a sustained, deep, obscure sound — acoustically associated with depth, continuity, and obscurity rather than sharpness and definition

Three properties of غ [ɣ] are particularly relevant to its semantic matrix:

- First, it is produced *deep in the posterior vocal tract*, deeper than almost any other Arabic consonant. This posterior depth is not merely a phonetic detail: it is physically experienced as interiority, as sound produced from within rather than projected outward.
- Second, it is *turbulent but non-explosive*, the airstream is continuous and uninterrupted, never reaching the complete closure and abrupt release that characterizes ق [q]. Where

[q] produces a discrete acoustic event, a boundary, [ɣ] produces a sustained acoustic process, a kind of engulfing.

- Third, its *acoustic obscurity*; the turbulent voiced friction of [ɣ] lacks the crisp definition of [q] release. It is a sound that does not delineate clearly but rather envelops, phonetically performing the very semantic content it encodes.

The contrast with ق [q] can now be stated with full precision:

Table 5.4 Contrast between ق [q] and غ [ɣ]

Feature	ق [q]	غ [ɣ]
Place	Uvular	Uvular
Manner	Stop — complete closure + abrupt release	Fricative — continuous turbulent flow
Voicing	Voiceless	Voiced
Acoustic quality	Sharp, discrete, outward burst	Deep, continuous, obscure resonance
Articulatory experience	Abrupt outward force	Sustained inward depth
Semantic matrix	Decisive boundary-creation	Engulfment, concealment, inward recession

This contrast shows strong evidence for articulatory iconic motivation: *same place of articulation, systematically opposite manner and acoustic properties, systematically opposite semantic fields*. This systematic double opposition, phonetic and semantic, is consistent with articulatory iconic motivation and difficult to account for under the arbitrariness thesis.

The opposition is not merely a collection of parallel observations. It is a structured iconic system. The shared uvular place of articulation provides the common domain; the contrasting manner and voicing encode opposite operations within that domain. The relationship between ق [q] and غ [ɣ] is one of motivated opposition, not arbitrary difference: their phonetic contrast iconically encodes their semantic contrast. This is the framework's strongest evidence: not just that individual consonants are motivated, but that their *contrasts are motivated*. *It is a structurally iconic opposition*.

5.3.2 The [ɣ] Semantic Matrix

From this articulatory profile we derive the following semantic matrix for [ɣ]: *Progressive engulfment into depth or darkness — the gradual disappearance of an entity into an obscuring, enveloping medium*. It contrasts precisely with the ق [q] matrix: where [q] creates a boundary *from outside* — severing, demarcating, deciding — [ɣ] dissolves a boundary *from within* — swallowing, submerging, concealing. The same uvular place of articulation generates opposite semantic operations through the contrast in manner.

5.3.3 Lexical Evidence of the [y] Matrix

Table 5.5 [y] Matrix Physical domain

Root	Meaning	Matrix instantiation
غرق	to drown, sink	engulfment by water
غطس	to dive, submerge	voluntary entry into engulfing depth
غاص	to dive deep	descent into obscuring medium
غمر	to flood, overwhelm	medium expanding to engulf
غطى	to cover, conceal	imposing a layer of concealment

Table 5.6 [y] Matrix Atmospheric/cosmic domain

Root	Meaning	Matrix instantiation
غاب	to be absent, disappear	paradigm case — withdrawal into invisibility
غرب	to set (sun)	disappearance below horizon
غيم/غام	to be overcast	sky engulfed by cloud
غسق	to grow dark	light swallowed by darkness

Table 5.7 [y] Matrix Psychological/social domain

Root	Meaning	Matrix instantiation
غفل	to be heedless	attention swallowed by distraction
غم	grief, suffocation	self-engulfed by overwhelming internal state
غرّ	to deceive	truth concealed within engulfing illusion
غريب	strange, foreign	one who has disappeared from their roots
غربة	Exile	disappearance from one's origin

The matrix of progressive engulfment into depth generates meanings as varied as physical drowning, solar setting, psychological distraction, and social alienation — all instantiating the same abstract schema: *an entity progressively disappearing into an enveloping medium* ⁵.

⁵ Not all غ-initial roots, however, seem to instantiate the engulfment matrix: غنى (to sing/be wealthy), غلا (to boil/be expensive), and غضب (to be angry) resist straightforward iconic motivation within the proposed

5.4 The ‘Cosmic’ Opposition of شرق *farq* vs. غرب *yarb*

Further evidence for the structured iconic opposition between غ [ɣ] and its contrasting consonants emerges from the cosmological pair شرق (*farq* East/sunrise) and غرب (*yarb* West/sunset). The two nouns present a consonantal architecture that is, we argue, iconically motivated at multiple positions within the root:

C1 [ɣ] vs. [q] — The Primary Iconic Opposition

- The initial consonant of غرب – [ɣ] – instantiates the engulfment matrix precisely: the setting sun disappears progressively below the horizon, swallowed by darkness in a continuous, gradual movement. The root √**y-r-b** generates a semantically coherent field entirely consistent with this matrix: غريب [*yarīb*] (strange, foreign — one who has disappeared from their roots), غربة [*yurba*] (exile — disappearance from one's origin), غرق [*yariqa*] (to drown — engulfment by water). [maɣrib] مغرب, carrying the {m-} nominalizing prefix analyzed in Section 3.6, is the reified spatial instantiation of engulfment: the place and time where the sun is swallowed into darkness.
-
- The initial consonant of شرق [ʃ], the palato-alveolar fricative, instantiates a contrasting matrix of *centrifugal dispersal*: the rising sun explodes outward, radiating light simultaneously in all directions. The مشرق [*maʃriq*], is correspondingly the reified place of radial explosion – where light erupts centrifugally across the sky. The contrast between غ [ɣ] and ش [ʃ] – engulfment vs. centrifugal dispersal – reinforces the broader consonantal opposition established in 5.4: two phonetically distinct consonants encoding semantically opposite operations on the same cosmic phenomenon.

C2 [r] — The Shared Movement

Both words share the medial consonant ر [r], the alveolar trill whose dynamic rolling articulation plausibly encodes continuous movement through space. Independent evidence supports this reading: roots such as جرى [*dʒarā*] (to flow/run), مرّ [*marra*] (to pass), دار [*dāra*] (to revolve), and فرّ [*farra*] (to flee) consistently associate ر [r] with dynamic spatial movement across diverse semantic domains. If this reading holds, the shared ر [r] in the *farq/yarb* pair reflects what sunset and sunrise have in common, the sun's uninterrupted daily journey, while the contrasting C1 consonants encode the opposite natures of that journey's beginning and end.

C3 [q] vs. [b] — An Asymmetric Picture

The final consonants present a more complex picture that we treat with appropriate caution. The final ق [q] of شرق [*ʃarq*] is strongly motivated within the framework: roots such as فلق [*falaq*] (to cleave), شقّ [*ʃaqqā*] (to split open), خرق [*ḫaraqa*] (to pierce), and برق [*baraqa*] (to flash) consistently associate final ق [q] with forceful, decisive action at the endpoint of a

schema. The matrix is therefore presented as a robust tendency rather than an exceptionless rule, consistent with the gradient model of motivation outlined in Section 2.1.

process. Applied to **شرق**, this reading is precise: the rising sun crosses the horizon as a sudden, decisive boundary event, the sun cleaving through the threshold between night and day.

The final **ب** [b] of **غرب** [yarb] suggests the gentle closing of the day, consistent with the bilabial closure, though independent lexical evidence for final **ب** [b] as a closure marker is thinner than for final **ق** [q], and we acknowledge this asymmetry honestly. The root **شرب** f-r-b ‘to drink’, however, whose act of drinking concludes with bilabial closure, offers a suggestive parallel, though not conclusive independent evidence. The reading is internally coherent within the *fāraq/yarb* architecture but requires further systematic investigation across the lexicon before it can be stated as a confirmed matrix property. However, tentatively, **شرق** [fāraq] encodes centrifugal explosion in **ش**/f, through continuous movement in **ر**/r, and across a decisive boundary **ق**/q, while **غرب** [yarb] encodes progressive engulfment with **غ**/y, through the same continuous movement in **ر**/r but arriving at gentle closure with **ب**/b. The **شرق/غرب** pair is presented not as proof of a fully developed three-consonant iconic system, but as a compelling illustration of consonantal iconicity operating beyond C1, and as confirmation of the structured opposition between [y] and [f] as contrasting consonants.

5.5 Brief Extension — **ف** [f] and **ش** [ʃ]

The **ق/غ** (q/ġ) minimal pair constitutes the primary evidence for the consonantal matrix system in this article. Brief mention of two further consonants, **ف**/f (f and ش) suggests that the system extends beyond the uvular pair and that contrasting phonetic gestures consistently encode contrasting semantic matrices:

ف [f] – the voiceless labiodental fricative – involves air escaping in a continuous directed stream through a narrow channel between the upper teeth and the lower lip. Its matrix is:

Outward directed release – the rupture of containment, releasing what was inside outward

Instantiated in roots such as **فتح** (fataha, to open), **فرق** (faraqa, to separate), **فلق** (falaqa, to split open), **فجر** (to burst open), **فرّ** (farra, to flee/escape), the schema is identical: something contained breaks open and releases outward in a specific direction.

ش [ʃ] – the palato-alveolar fricative – involves the tongue spread wide, producing turbulent air that fans out across a broad channel in multiple directions simultaneously. Its matrix is: *Centrifugal dispersal – spreading, ramifying, scattering in multiple directions simultaneously*

Instantiated in roots such as **شقّ** (shaqqa, to crack/split), **شتّت** (fattata, to scatter), **شاع** (šāʿa, to spread/circulate), **شعّ** (šaʿʿa, to radiate light), **شرح** (sharaḥa, to explain/spread out).

The contrast between **ف** and **ش** is itself iconically motivated: **ف** releases outward in a *single directed stream* – one direction, one trajectory – while **ش** disperses in *multiple simultaneous directions* – radial, centrifugal, non-directed.

5.6 Positioning Within the Existing Literature

The consonantal matrix system presented in this section engages with and goes beyond two major predecessors:

- **Ibn Jinnī's *Al-ishtiqāq al-akbar*** correctly intuited that consonants carry semantic weight independently of their root position; his permutational approach anticipated the matrix concept. However, his framework is limited to demonstrating shared semantic fields across permuted roots, without proposing a phonetically grounded explanation about specific consonants clustering around specific semantic fields. The present framework provides that explanation: the articulatory properties of the consonant iconically pre-figure its semantic matrix.
- **Bohas's *Matrix-Etymon theory*** advances beyond Ibn Jinnī by identifying specific consonant pairs as semantic determinants, a significant step toward the present analysis. However, Bohas (1997) grounds his matrices in abstract phonological features rather than articulatory iconicity, which limits the cognitive and cross-linguistic motivation of his account. Furthermore, Bohas does not address the transparent correspondence between lexical roots and grammatical particles – the domain examined in Section 4. The present framework is therefore both more concrete in its grounding, articulatory gesture rather than abstract feature, and more extensive in its scope, reaching from the sub-root level through the lexicon to the grammatical particle system.

The present study proposes a theoretical framework of articulatory iconic motivation in Arabic morphology, distinct from both McCarthy's formal phonological account (1981) and Bohas's (1997) abstract feature matrices. While McCarthy's framework correctly describes the architectural conditions that preserve iconic motivation, and Bohas correctly identifies sub-root semantic organization, neither framework accounts for the systematic relationship between individual consonants' articulatory properties and their associated semantic fields. The present research proposes that this relationship is iconically motivated, grounded in the embodied experience of sound production, and constitutes a third, complementary level of analysis.

6. Theoretical Implications and Conclusion

6.1 Against the Arbitrariness Principle

The evidence presented across Sections 3, 4, and 5 converges on a single conclusion: the relationship between phonological form and meaning in Arabic is systematically non-arbitrary at multiple simultaneous levels. The verb form paradigm is diagrammatically iconic: its phonological operations enact the semantic operations they encode. The {m-} prefix nominalization system is grounded in articulatory iconicity whose recurrence cannot be explained by convention. The cases examined in Section 4 demonstrate that Arabic grammatical particles show transparent motivated correspondence with related lexical roots — the same articulatory iconicity that organizes the root lexicon extends into the grammatical system. The

consonantal matrix system shows that individual consonants, through their articulatory properties, iconically pre-figure the semantic fields of the roots they initiate.

Taken together, these four domains of evidence constitute a challenge to the Saussurean arbitrariness thesis that operates at a different level from previous critiques. Earlier challenges – from Jespersen's sound symbolism observations to Hinton, Nichols and Ohala's cross-linguistic survey – have documented iconic phenomena at the periphery of linguistic systems: expressive vocabulary, onomatopoeia, ideophones. The present study demonstrates iconic motivation at the **core** of a morphological system, in its most productive derivational categories, its most fundamental grammatical particles, and its most basic organizational principle.

The arbitrariness principle is not a universal property of human language but a **default tendency** that specific morphological architectures systematically override. by keeping consonantal roots visible, stable, and analytically separable across derivation, root-and-pattern morphology creates the structural conditions under which iconic motivation can be preserved, elaborated, and institutionalized as the organizing principle of the entire morphological system.

6.2 Beyond Bohas — Articulatory Grounding and Grammatical Extension

The present framework advances beyond Bohas's Matrix-Etymon theory in two respects:

- First, by grounding the consonantal matrix system in **articulatory iconicity** rather than abstract phonological features, we provide a cognitively and physiologically motivated account of why specific consonants cluster around specific semantic fields. The answer is not that certain feature combinations happen to be associated with certain meanings – it is that the physical act of producing a consonant **enacts** the conceptual primitive it encodes. The uvular stop [q] physically performs boundary-creation; the uvular fricative [χ] physically performs engulfment. The motivation is embodied before it is cognitive, and cognitive before it is conventional.
- Second, by extending the analysis into the grammatical particle system, we show that consonantal iconicity is not confined to the root lexicon. Arabic grammatical particles show transparent motivated correspondence with related lexical roots — the iconic motivation is synchronically active and observable, not merely a historical residue. This extension, to our knowledge, has not previously been proposed in the literature and constitutes the most original theoretical contribution of the present study.

6.3 Why Arabic?

A question that naturally arises from this analysis is: Why does Arabic, more than other languages, preserve and display iconic motivation so systematically? Four converging factors suggest an answer:

- **The non-concatenative architecture:** root-and-pattern morphology keeps consonants visible and analytically separable, protecting iconic motivation from the phonological erosion that buries it in concatenative languages.

- **The consonantal writing system:** Arabic script foregrounds consonants cognitively – vowels are secondary – reinforcing the salience of consonantal matrices in both production and acquisition.
- **The oral-poetic tradition:** pre-Islamic Arabic poetry created intense selective pressure favouring lexical choices that exploited the iconic resources of the consonantal system, refining and reinforcing the iconic architecture over centuries of oral transmission.
- **The Quranic stabilization:** the requirement of precise Quranic recitation – preserved in the science of *tajwīd* across fifteen centuries – created an unprecedented phonological conservatism that maintained the distinctions on which the iconic system depends.

These four factors do not explain the origin of Arabic's iconic architecture. The question remains open, but they explain its remarkable **preservation and stability** relative to other languages, including other Semitic languages that share Proto-Semitic roots but have undergone phonological erosion.

6.4 Implications for Arabic Pedagogy

The iconic architecture described in this study has direct implications for Arabic language teaching that deserve brief mention. If the Arabic morphological system is iconically organized rather than arbitrarily structured, then learners who internalize the iconic motivation of the verb form paradigm and the consonantal matrices possess a *generative key* to the lexicon, one that makes the meaning of unfamiliar forms derivable rather than merely guessable. A learner who understands that gemination encodes intensification, that the {m-} prefix concretizes verbal processes into nominal entities, and that [q]-initial roots cluster around boundary-creation has not merely memorized a set of rules, but internalized an iconic system that generates correct predictions across thousands of derived forms. This generativity is the most practical consequence of the iconic architecture, and its most powerful argument against the pedagogical tradition that presents Arabic morphology as an arbitrary system to be acquired through rote memorization.

6.5 Opening Toward the Book

The present article has necessarily been selective. The consonantal matrix system has been illustrated through two primary cases – ق and غ – with brief extensions to [f] ف and [ʃ] ش. The motivated correspondence between lexical items and grammatical particles has been limited to a few case studies. The verb form paradigm has been analysed for its most transparent iconic relationships. The full scope of the iconic architecture of Arabic – covering all twenty-eight consonants, the complete verb form inventory, the full particle and preposition system, and the comparative Semitic dimension – requires the book-length treatment of which this article is the first instalment.

Several directions for future investigation are particularly pressing. First, *a systematic survey* of all Arabic consonants, testing the matrix hypothesis across the full consonantal inventory and documenting coverage and outliers honestly. Second, *a comparative Semitic*

analysis – examining whether the same matrices are recoverable in Hebrew, Aramaic, and Akkadian cognate roots, which would locate the iconic motivation at the Proto-Semitic level and rule out Arabic-specific conventionalization as an alternative explanation. Third, *a dialectal dimension* – exploring how the iconic architecture has evolved in spoken Arabic varieties

6.6 Conclusion

The article has argued that the Arabic linguistic sign is not arbitrary but motivated at multiple levels of the linguistic system. The verb form paradigm is diagrammatically iconic, its phonological operations enacting their semantic content. The {m-} prefix nominalization system is grounded in the articulatory iconicity of the bilabial nasal — an iconicity with cross-linguistic resonance, as evidenced by the near-universal presence of [m] in words for ‘mother’ across unrelated language families. Arabic grammatical particles show transparent formal and semantic correspondence with related lexical roots. And individual consonants, through their articulatory properties, iconically pre-figure the semantic fields of the roots they initiate.

The most original contribution of the present framework is the consonantal matrix system, the demonstration that individual consonants carry stable semantic matrices grounded in their articulatory properties, and that contrastive phonetic gestures encode contrastive semantic matrices in a structured iconic opposition. The غ/ق (ȝ/q) minimal pair is the clearest evidence: same place of articulation, systematically opposite manner and acoustic properties, systematically opposite semantic fields. This is not coincidence. It is motivated opposition.

The implications extend beyond Arabic linguistics. If a natural language can sustain phonosemantic iconicity as a systemic architectural property, at the level of the morphological paradigm, the derivational system, and the grammatical particle inventory simultaneously, then the Saussurean arbitrariness thesis cannot be maintained as a universal principle. It must be reconceived as a default tendency, one that specific morphological architectures evidently prevail, particularly non-concatenative ones.

The present study advances a theoretical framework rather than a fully established theory. The consonantal matrix system requires further systematic investigation across the full Arabic consonantal inventory, comparative Semitic evidence, and psycholinguistic testing. These are directions for future research. The study has established the framework's analytical coherence and its empirical grounding in four interconnected domains of Arabic morphology and grammar. It offers systematic evidence that the motivated sign is not merely a theoretical possibility but an empirically demonstrable property of Arabic morphology and grammar. *The motivated sign exists. Arabic is its clearest demonstration.*

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